Cleveland Clinic

Gestational Diabetes

Gestational diabetes is high blood sugar during pregnancy. Eating healthy, well-balanced foods and getting exercise can usually keep it well managed. But sometimes, insulin is necessary to help you manage blood sugar levels. If left untreated, it can cause health problems for both you and the fetus.

Overview

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Gestational diabetes is a type of diabetes that happens during pregnancy. It may not cause any symptoms. You'll have a screening for gestational diabetes during pregnancy.

What is gestational diabetes?

Gestational diabetes (GD or GDM) is a type of diabetes that develops exclusively in pregnancy when blood sugar levels get too high (hyperglycemia). It happens when the hormones from the placenta block your ability to use or make insulin. Insulin helps your body maintain the right amount of glucose in your blood. Too much glucose in your blood can lead to pregnancy complications. GD usually appears during the middle of pregnancy, between 24 and 28 weeks. Your pregnancy care provider will order a blood test to check for gestational diabetes.

Developing GD doesn't mean you already had diabetes before you got pregnant. The condition appears *because of* pregnancy. People with Type 1 and Type 2 diabetes before pregnancy have their own, separate challenges when they become pregnant.

Fortunately, gestational diabetes is well understood, and healthcare providers are usually able to help you manage the condition with small lifestyle and dietary changes. Most people don't experience serious complications from gestational diabetes and deliver healthy babies.

What happens if you get gestational diabetes?

If you have gestational diabetes, your pregnancy care provider will have you visit a nutritionist who specializes in gestational diabetes. At this appointment, you'll talk about how certain foods typically increase blood sugar levels and how to make sure your meals and snacks contain the right types and amounts of food.

They'll also talk to you about checking or testing your blood sugar at home, and what your levels should be after meals.

Finally, your obstetrician or nutritionist will discuss a gestational diabetes management plan. People with diabetes typically have more prenatal visits to check on fetal growth, monitor their weight gain and discuss how well they're managing their blood sugar.

Rate of gestational diabetes U.S. and worldwide

The rate of gestational diabetes in the United States is rising. According to the U.S. Centers for Disease Control and Prevention (CDC), about 8% to 10% of pregnant women will develop GD.

The rate of gestational diabetes worldwide, on average, is between 14% and 17%. Other factors can contribute to rates being higher, such as age, race/ethnicity, access to prenatal care and geography.

Symptoms and Causes

What causes gestational diabetes?

Gestational diabetes comes from hormonal changes and the way your body converts food into energy.

A hormone called insulin breaks down the glucose (sugar) from food and delivers it to your cells. Insulin keeps the level of glucose in your blood at a healthy level. But if insulin doesn't work right or you don't have enough of it, sugar builds up in your blood and leads to diabetes.

During pregnancy, hormones from the placenta can interfere with the way insulin works. It may not regulate your blood sugar levels like it's supposed to, which can lead to gestational diabetes.

Who is at risk for gestational diabetes?

Anyone can get gestational diabetes during pregnancy. But certain factors can increase your risk. They include:

- Heart disease.
- High blood pressure.
- Having obesity or overweight before pregnancy.
- Personal history of GD (in previous pregnancies).
- Family history of Type 2 diabetes.
- Polyendocrine metabolic ovarian syndrome (PMOS).
- Advanced maternal age.
- Prediabetes (history of higher-than-normal blood glucose).

People over the age of 25 who are of South and East Asian, Hispanic, Native American or Pacific Island descent are also at a higher risk.

What are the symptoms of gestational diabetes?

There are usually no obvious warning signs of gestational diabetes. Symptoms are mild and often go unnoticed until your pregnancy care provider tests you for gestational diabetes.

If you do have symptoms, they may include:

- Frequent urination.
- Excessive thirst.
- Tiredness.
- Nausea.

What are complications of unmanaged gestational diabetes?

Being unable to keep your blood sugar levels in a healthy range can lead to complications for both you and the fetus.

Gestational diabetes increases your risk of:

- C-section (if the fetus gets too big).
- Preeclampsia (high blood pressure during pregnancy).

People with gestational diabetes are slightly more likely to have Type 2 diabetes later in life.

How does gestational diabetes affect my baby?

If you have gestational diabetes, your baby's more at risk for:

- Increased birth weight (fetal macrosomia).
- Breathing problems at birth.
- Hypoglycemia. This can cause seizures in newborns.
- Obesity.
- Premature birth.
- Type 2 diabetes later in life.

Diagnosis and Tests

When does a healthcare provider test for gestational diabetes?

Your healthcare provider tests for gestational diabetes around weeks 24 to 28 of your pregnancy. Most healthcare experts believe this is the best time to get the most accurate results. But your provider may screen you for gestational diabetes a little earlier, especially if you have risk factors for the condition.

How is gestational diabetes diagnosed?

Your healthcare provider will test your blood sugar during pregnancy with a blood test. Often, the first test is a "screening," which means it identifies people who are more likely to have gestational diabetes. If you don't pass the screening, your provider will order a second blood test which is more comprehensive.

- **Glucose challenge test:** This test is sometimes called a glucose screening test or a one-hour glucose tolerance test. You drink a sweet liquid and wait in a waiting room for one hour. After one hour, a healthcare provider will draw a blood sample from your arm. If your blood sugar is too high, your healthcare provider will order a glucose tolerance test.
- **Glucose tolerance test:** This test is sometimes called a two-hour or three-hour glucose test. This test is only done if your challenge test results are irregular. You fast (don't eat for eight hours) before the tolerance test. Your healthcare provider draws your blood before and at one-, two- and/or three-hour intervals after you drink a sweet liquid. The tolerance test can confirm a diagnosis of gestational diabetes.

Management and Treatment

How is gestational diabetes managed?

If you have gestational diabetes, you'll need more frequent checkups during your pregnancy. Your pregnancy care provider will want to:

- Monitor the growth of the fetus. This typically involves having a few extra ultrasounds where they'll make sure the fetus isn't getting too large.
- Review your blood sugar ranges. Most of the time, this involves discussing how often your blood sugar is high or low and looking at general trends in your blood sugar levels.

To keep track of your blood sugar at home, you'll need a tool called a glucose meter. You'll need the monitor itself, as well as needles (or lancets) and test strips. Taking your blood sugar involves pricking the tip of your finger with the lancet and then wiping the blood on a test strip. Then, you insert the test strip into the meter. After a few seconds, the device will display a number. This is your blood sugar level. Your nutritionist or diabetes educator will go over how to use the device. It may take a few days to get the hang of it, but your provider is always available to help you if you need it.

You'll have to record your meals and blood sugar readings on paper, on an app or whatever way your provider wants you to. Then, your pregnancy care provider will review your readings at regular intervals (often weekly or biweekly). Sometimes, your provider will need to adjust your food (like decreasing the amount of carbs you eat) to keep your glucose levels well managed. The combination of tracking your blood sugar levels and eating diabetes-friendly food is usually enough to manage gestational diabetes.

Some people need medication such as insulin to manage gestational diabetes. This usually involves injecting insulin into your stomach, thigh or buttocks every day or multiple times a day. How frequently you need to inject insulin depends on many factors. Your healthcare provider will teach you how to inject insulin, when to take it and how much to take. If you need to use insulin to manage diabetes, it's important to take it exactly as your provider prescribes.

Blood sugar levels

You typically record your glucose levels right when you wake up (before eating anything) and then about one hour after each meal. For example, you may take your blood sugar upon waking up at 6:15 a.m. Then, if you eat breakfast at 7 a.m., you'd take your blood sugar level around 8 a.m. You record the numbers in whatever way you and your provider have agreed on.

The American College of Obstetricians and Gynecologists typically recommends the following ranges for blood glucose levels:

- Before a meal: 95 milligrams (mg)/dL or less. (Mg/dL stands for milligrams per decilitre.)
- One hour after a meal: 140 mg/dL or less.
- Two hours after a meal: 120 mg/dL or less.

As always, everyone is unique, and these are only guidelines. Your pregnancy care provider will discuss what *your* blood sugar range should be.

What shouldn't you eat with gestational diabetes?

You can help manage gestational diabetes by eating nutritious foods that don't cause your blood sugar to rise to unsafe levels. Try to:

- Avoid processed foods and sugary drinks.
- Choose a healthy balance of proteins, carbohydrates, fiber and fat for each meal.
- Eat smaller meals more often.
- Schedule your meals at the same time each day.

Some people with gestational diabetes find that small tweaks to what they usually eat are all that's necessary to maintain normal blood sugar levels. Still, some find that they have to make more changes. It's

important to keep in mind that some people may be able to eat certain foods without a large impact on blood sugar, but you may not be able to. Remind yourself that everyone is different and unique.

Does drinking water lower blood sugar with gestational diabetes?

Hydration is important during pregnancy, especially if you have gestational diabetes. But there are no studies that show drinking water alone will directly lower blood sugar.

Staying hydrated does help your body regulate your blood sugar, though. It helps dilute your blood, which makes it easier for your kidneys to get rid of excess sugar. Water can also help your body use insulin more effectively, which also contributes to regulating blood sugar.

How can I exercise safely with gestational diabetes?

Exercise helps your body use more glucose, which may lower your blood sugar. If you have gestational diabetes, talk to your healthcare provider about creating a safe exercise plan. Your exercise plan should be unique to your overall health and your fitness level before pregnancy.

How can you reverse gestational diabetes while pregnant?

You can't reverse gestational diabetes once you have it. Your provider will test you for diabetes after your baby is born (usually between six and 12 weeks postpartum) to see if you still have it.

Outlook / Prognosis

How worried should I be about gestational diabetes?

Gestational diabetes is a common condition and healthcare providers have a good idea of how best to manage and treat it. You'll still have a healthy pregnancy and a healthy baby if you have gestational diabetes. Work with your healthcare provider to make sure you understand your treatment plan and how you can keep your blood sugar levels healthy.

Take time to understand the possible complications of not managing gestational diabetes. Your baby has a very good chance of being born healthy, but you must take steps to manage the condition. If your blood sugar levels are high several readings in a row, don't wait to contact your provider. Let them know that your blood sugar levels are repeatedly high so they can adjust your foods or medication and help you. Gestational diabetes is manageable, but there's a level of responsibility you must take to ensure your pregnancy is healthy.

Will gestational diabetes go away after pregnancy?

Your blood sugar levels should come down after you give birth, when your hormone levels return to normal. Your pregnancy care provider will test you for gestational diabetes after your baby is born to confirm it's gone (usually around six to 12 weeks postpartum).

But about 50% of people with gestational diabetes develop Type 2 diabetes later in life. Eating the right foods for your body and getting physical activity can help lower your risk. Your healthcare provider may recommend blood glucose tests every few years to watch for diabetes, especially if you have one or more risk factors.

Does having gestational diabetes make a pregnancy high risk?

Yes, having gestational diabetes may make your pregnancy high risk. Healthcare providers consider a pregnancy high risk when either you or the fetus (or both) has health conditions that increase your chances

of having a pregnancy complication.

Will my baby be healthy if I have gestational diabetes?

Yes. Most babies born are born healthy. There are some steps you can take to manage gestational diabetes during pregnancy to give your child the best start in life. Attending all your prenatal appointments and managing diabetes the best you can during pregnancy are the two best things you can do.

Prevention

How can gestational diabetes be prevented?

It's not entirely preventable, but you can take steps to reduce your risk. Eating balanced and nutritious meals and getting regular exercise before and during your pregnancy are the best ways to reduce your risk of developing gestational diabetes.

Living With

What can I do to make living with gestational diabetes easier?

Make diabetes management part of your daily routine. Create a schedule and stick to it. Try to:

- Check your blood glucose levels at the same time each day.
- Choose three days each week to get 30 minutes of light exercise.
- Plan small, balanced meals ahead of time.
- Talk with your healthcare provider or a diabetes educator about other tips for daily diabetes management.

When should I see my healthcare provider if I have gestational diabetes?

Even if you're being careful to manage your condition, there may be situations where you need to call your pregnancy care provider. Contact your provider if you have gestational diabetes and:

- **You're having trouble managing blood sugar levels**. This means your blood sugar levels are higher than the range your provider gave you for several readings in a row. They may want to adjust your diabetes management plan.
- **Your blood sugar is consistently low**. Having low blood sugar can be a bad thing, too. Your provider may have ideas to help your blood sugar stay in a healthier range.
- **You have an illness that prevents you from following your management plan**. For example, you may have food poisoning or be vomiting for another reason. Being unable to eat will affect your blood sugar levels.

Additional Common Questions

Can you eat healthy and still get gestational diabetes?

Yes, you can get gestational diabetes even if you ate healthy before or during pregnancy. There are several other factors that go into your risk of getting diabetes. Things like hormones and genetics play a major role and those factors are beyond your control.

But if you have gestational diabetes, eating healthy, nutritious foods is one of the best ways to manage the condition throughout your pregnancy.

A note from Cleveland Clinic

Your pregnancy care provider just told you that you “failed” your glucose test and that you have gestational diabetes. A million questions may be crossing your mind. Did I cause this? Do I need to prick my finger the rest of my pregnancy? Will my baby be born healthy? These questions (and others) are normal to have. Fortunately, healthcare providers have a good understanding of gestational diabetes and how to help you manage it.

Most people can manage gestational diabetes with small changes to what they eat. A nutritionist or diabetes educator can help you with that. Still, some people need a medication called insulin to help keep their blood sugar in check. In most cases, well-managed gestational diabetes doesn’t have a major impact on your pregnancy. But you have to play an active role in managing the condition to make sure your pregnancy and your baby are healthy.

Experts You Can Trust

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References

Cleveland Clinic’s health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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